Round 2 - “Growing Your Outpost”

It is the second trading round, and your final opportunity to reach the threshold goal of a net PnL of 200,000 XIRECs or more before the leaderboard resets for Phase 2. These first 2 rounds act as qualifiers for the final mission. Trading activity has accelerated significantly since your arrival. With you and the other outposts actively trading ***Ash-Coated Osmium*** and ***Intarian Pepper Root***, the market has become increasingly competitive and dynamic.

In this second and final trading round on Intara, you will continue trading `ASH_COATED_OSMIUM` and `INTARIAN_PEPPER_ROOT`. This time, however, you have the opportunity to gain access to additional market volume. To compete for this increased capacity, you must incorporate a ***Market Access Fee*** bid into your Python program.

Of course, you should also analyze your previous round’s performance and refine your algorithm accordingly.

Additionally, XIREN has provided a ***50,000 XIRECs investment budget*** for you to allocate across three growth pillars in order to accelerate the development of your outpost. You must decide how to distribute this budget strategically to maximize your profit once the trading round closes.

Round Objective

Optimize your Python program to trade `ASH_COATED_OSMIUM` and `INTARIAN_PEPPER_ROOT`, and incorporate a ***Market Access Fee*** to potentially gain access to additional market volume.

In addition to refining your trading algorithm, ***allocate your 50,000 XIRECs investment budget*** across the three growth pillars to strengthen your outpost’s performance.

Algorithmic trading challenge: “limited Market Access”

Wiki_ROUND_2_data.zip 689.3 KiB

The products `INTARIAN_PEPPER_ROOT` and `ASH_COATED_OSMIUM` are the same, but the challenge now primarily lies in deciding how much to bid for extra market access, as well as refining your algorithm. The position limits (see the Position Limits page for extra context and troubleshooting) are again

- `ASH_COATED_OSMIUM`: 80
- `INTARIAN_PEPPER_ROOT`: 80

In this round, you can bid for 25% more quotes in the order book. The volumes and prices of these quotes fit perfectly in the distribution of the already available quotes. A simple example:

> 📖 **Example Extra Market Access**
>
> Order book for participants with no extra market access:
> (ask, 10 volume, $9)
> (ask, 10 volume, $7)
> (bid, 10 volume, $5)
> (bid, 5 volume, $4)
>
> Order book for participants with *extra* market access:
> (ask, 10 volume, $9)
> (ask, 5 volume, $8)   <--- extra flow to trade against
> (ask, 10 volume, $7)
> (bid, 10 volume, $5)
> (bid, 5 volume, $4)

You bid for extra market access by incorporating a `bid()` function inside your `class Trader` implementation:

```
class Trader:
    def bid(self):
        return 15

    def run(self, state: TradingState):
        (Implementation)
```

The Market Access Fee (MAF) is a *one-time fee* at the start of Round 2 paid *only* if your bid is accepted. It only determines who gets extra market access, and is not used in the simulation dynamics whatsoever. The top 50% of bids across all participants are accepted.

🔨 **Example Bidding Mechanism**

Bids: [10, 20, 15, 19, 21, 34]
Accepted: [No, Yes, No, No, Yes, Yes]
Explanation: the median of the bids is 19.5, so all bids higher (20, 21, 34) are accepted → these participants get extra market access flow while paying the price they bid, and all bids below 19.5 are rejected (and these participants do *not* pay the fee).

The accepted bids are subtracted from Round 2 profits to compute the final PnL. To be explicit,

> ℹ️ For those with full market access (i.e. those in the top 50% of bids), `profit = profit from round 2 - bid for getting full market access`.
>
> For those with no full market access, `profit = profit from round 2`.

The MAF is unique to Round 2, and does not apply to any other round; any `bid()` function in Rounds 1,3,4,5 is ignored. It is also ignored during testing of round 2, since bids are only compared on our end when the final simulation of Round 2 starts. In that sense, it's a "blind auction" for extra flow.

During testing of round 2, the default set of quotes you interact with is 80% of all quotes we generated (i.e., no extra market access). This 80% has been slightly randomized for every submission to reflect real-world conditions where not all patterns in trading behavior are up 100% of the time. While you could optimize the PnL by submitting the same file many times, this has very limited payoff and your effort is much better put into improving your algorithm ;).

Game theory

To get extra market access, you just need to be in the top 50% of bidders, not necessarily the highest bidder. Placing an extremely high bid will almost certainly yield full market access, but perhaps you could save (a lot of) XIRECs by bidding less while staying in the top 50% of bidders.

Manual trading challenge: “Invest & Expand”

You are expanding your outpost into a true market making firm with a budget of `50 000` XIRECs. You need to allocate this budget across three pillars:

- **Research**
- **Scale**
- **Speed**

You choose percentages for each pillar between 0–100%. Total allocation cannot exceed 100%. Your final PnL (Profit and Loss) score is:

> ℹ️ PnL = (Research × Scale × Speed) – Budget_Used

The pillars

Research determines how strong your trading edge is. It grows **logarithmically** from `0` (for `0` invested) to `200 000` (for `100` invested). The exact formula is `research(x) = 200_000 * np.log(1 + x) / np.log(1 + 100)`. Here, `np.log` is a python function from NumPy package for natural logarithm.

Scale determines how broadly you deploy your strategy across markets. It grows **linearly** from `0` (for `0` invested) to `7` (for `100` invested).

Speed determines how often you win the trades you target. It is **rank-based** across all players:

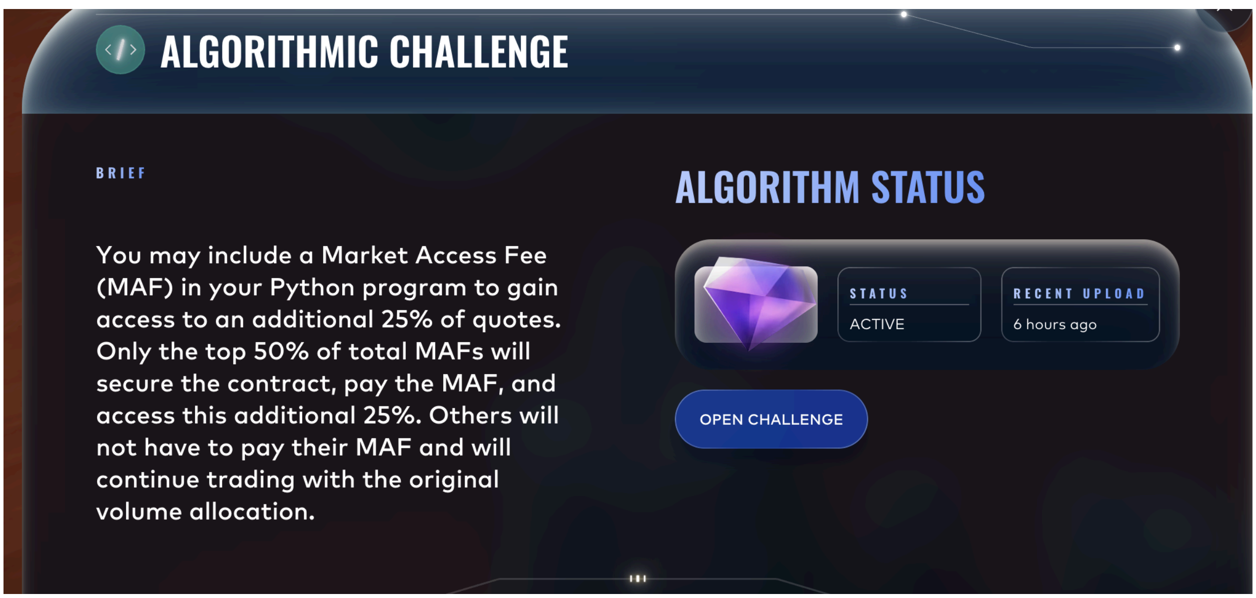
- Highest speed investment receives a `0.9` multiplier.
- Lowest receives `0.1`.
- Everyone in between is scaled linearly by rank, equal investments share the same rank.
- For example, if people invested `70, 70, 70, 50, 40, 40, 30`, they get the following ranks: `1, 1, 1, 4, 5, 5, 7`. First three players get `0.9` for hit rate multiplier, last player gets `0.1`, and everybody in between gets linearly scaled between top and bottom rank. Another example, if you have three players investing `95, 20, 10`, their ranks are `1, 2, 3`, and their hit rates are `0.9, 0.5, 0.1`.

Your Research, Scale, and Speed outcomes are multiplied together to form your gross PnL, after which the used part of your budget is deducted.

Every decision you make reflects a real trade-off faced by modern market makers: capital is finite, competition is relentless, and edge alone is never enough. Good luck!

Submit your orders

Choose the distribution of your budget by assigning percentages to the three pillars directly in the Manual Challenge Overview window and click the “Submit” button. You can re-submit new distributions until the end of the trading round. When the round ends, the last submitted distribution will be locked in and processed.



MANUAL CHALLENGE

INVEST & EXPAND

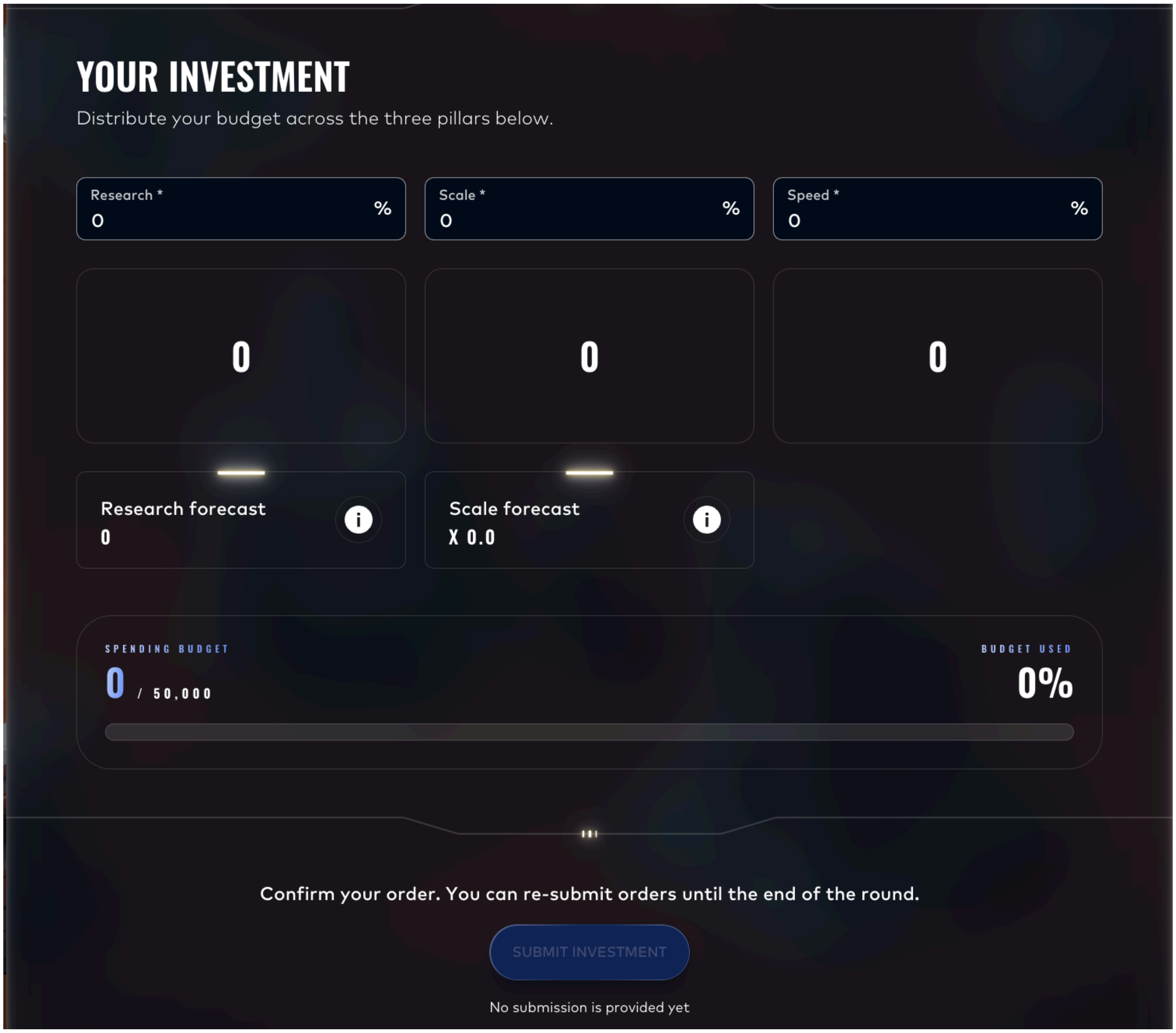
You've been granted a substantial budget to expand your trading outpost. You have 50,000 XIRECs at your disposal to invest across three growth pillars: Research, Scale, and Speed.

Investments in Research are directed toward quantitative research, data acquisition, machine learning infrastructure, backtesting systems, and related capabilities.

Budget allocated to Scale supports traders, engineers, offices, exchange access, and other factors that allow your operations to expand.

Investments in Speed are applied to technologies such as FPGA boards, colocation, RF towers, microwave links, and other latency-reducing infrastructure.

This is a one-time opportunity and does not affect your algorithmic trading activities. Enter your submission below and click Submit to confirm.


YOUR INVESTMENT
Distribute your budget across the three pillars below.
Research *
0
%
Scale *
0
%
Speed *
0
%
0
0
0
Research forecast
0
Scale forecast
X 0.0
SPENDING BUDGET
0
/ 50,000
BUDGET USED
0%
Confirm your order. You can re-submit orders until the end of the round.
SUBMIT INVESTMENT
No submission is provided yet

ROUND 2

Opportunity Valuation

SELECTED

Same products. New question. Okay. At least we're moving forward.

This round it stops being about what the goods are worth and starts being about what access is worth. To you. Specifically. Which is actually a more interesting question, if you think about it. Most people don't, but still.

There's a fee attached to the additional volume. Fine. Expected. But, a fee might also cost you more than the extra volume was ever going to give back. That's the line. The point where the opportunity stops being an opportunity and just becomes an expense with ambition.

I'd draw you a diagram but honestly the math is not that complicated once you stop being distracted by how exciting "additional volume" sounds. It's not that exciting. It's just volume.

Ask yourself: does the access still make sense once the fee is in the picture? Does it change how you price things? Does it shift where you want to position yourself? If any of those answers feel wobbly, you've got some recalculating to do.

Don't get excited. Get accurate. Those two things rarely happen at the same time.

Competing With Others

SELECTED

Right. So now it gets a little more interesting. Because this is not just about your math anymore. It is about everyone else's math. And what they think your math looks like. And what you think they think. Yeah. It gets like that.

To get full market access you need to outbid the median. Not the top. The median. Which means you are not trying to beat the most aggressive person in the room. You are trying to read the room. Slightly different skill. Obviously.

Every bid out there is basically someone's guess about what everyone else is going to do. Including yours. So the question is not just what access is worth to you. It is what you think it is worth to them, and how hard they are going to go for it.

Underestimate the field and you lose out on extra access entirely. Overestimate and you overpay for something that was never going to justify the cost. Both outcomes are avoidable. Both are also extremely common, for what it's worth.

So read the room. Conservative field or aggressive one? The answer changes where the median lands. And where the median lands changes everything.

You don't need to be right about everyone. Just close enough to land above the middle. That's it. Not complicated. Just easy to overthink.

Risk Appetite

SELECTED

Oh please. Don't be desperate. You were going to overbid by miles, weren't you?

Full market access goes to the top half. Not the top bid. The top half. Which means bidding extremely high doesn't make you smart. It just makes you someone who overpaid. Not a great look.

The more interesting move is figuring out how low you can go while still landing above the median. Every XIREC you don't spend on access is a XIREC that actually does something useful. Efficiency isn't boring. It's just underrated. Unlike overbidding, which is both boring and expensive.

The catch, obviously, is that the closer you bid to the threshold, the thinner your margin for error gets. Misjudge the field and you fall below the median. Access gone. Which, depending on your strategy, ranges from mildly annoying to a complete disaster.

So figure out how confident you actually are in your read of the field. Confident enough to bid close to the edge? Or do you need a little more breathing room? Neither is wrong. They're just different risk profiles.

Pick the one that matches your situation. Not the one that sounds braver than you feel right now.

DETAILED RESULTS

RICH OR BEHIND WENDYS

POSITION	XIREC
1139	183,520

ALGORITHMIC TRADING RESULTS

+ 95,524

ROUND RANKING 1629TH

MANUAL TRADING RESULTS

+ 87,995

ROUND RANKING 1ST

Previous Total	0
Round 1 Total	183,520
New Total PnL	**183,520**

MISSION PROGRESS

92%

TEAM GOAL

183,520

OF 200,000

PROSPERITY

2

CONSISTENCY IS KEY

CREW HONORS

10 BADGES UNLOCKED

See which badges your team unlocked and how.

VIEW ALL BADGES